Exploration: A Cyclodehydration Reaction

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Linear free energy relationships are supposed to be linear, and most of the time they are. However, when we have two or more competing pathways or two or more steps in a reaction mechanism, we might observe curved Hammett plots. In this document, we will explore an example.

Class Discussion: Curved Hammett Plots

Electrophilic aromatic substitution is a classic mechanism that you will be familiar with. In this experiment, Hart and Sedor investigated the mechanism when the reaction involves intramolecular ring closing.[1] Normally, the addition to the aromatic ring would be rate-determining (k_2 in Figure 1.) However, here we have that reaction occurring intramolecularly and that will provide for a much greater rate. Perhaps it will be faster than the formation of the carbocation electrophile, and we will see that first step being rate-determining.

The authors synthesized substituted versions of the starting material. The σ values for the substituents are added together to report a total σ value ($\sum \sigma$) for each polysubstituted reactant. The authors used σ in one region of the Hammett plot and σ^+ in another. They had their reasons, but they did recognize that this made little difference to the quality of the line fits. In this analysis, I will use only σ^+. The intermediate is a benzyl cation, the very definition of when to use σ^+. The substituent constants are presented in table 2 and the reactants with their $\sum \sigma^+$ values are described in table 1.

Reactant	X	Y	Z	$\sum \sigma$	$\sum \sigma^+$	k_{obs}
1	H	p-Cl	p-Cl	0.46	0.22	0.096
2	m-CF_3	H	H	0.43	0.43	0.052
3	m-Cl	H	H	0.37	0.37	0.100
4	H	p-Cl	H	0.23	0.11	0.30
5	m-OCH_3	H	H	0.12	0.12	0.74
6	H	H	H	0	0	1.00
7	m-CH_3	H	H	-0.07	-0.07	1.48
8	H	p-CH_3	H	-0.17	-0.31	3.27
9	H	p-OCH_3	H	-0.27	-0.78	0.56
10	H	p-CH_3	p-CH_3	-0.34	-0.62	5.80
11	H	p-CH_3	p-OCH_3	-0.44	-1.09	**0.176**
12	H	p-OCH_3	p-OCH_3	-0.54	-1.56	0.009

[1] "Mechanism of cyclodehydration of 2-phenyltriarylcarbinols", H. Hart, E.A. Sedor, *J. Am. Chem. Soc.*, **1967**, *89*, 2342-2347. https://doi.org/10.1021/ja00986a018

Figure 1: Cyclization via electrophilic aromatic substitution ⟵

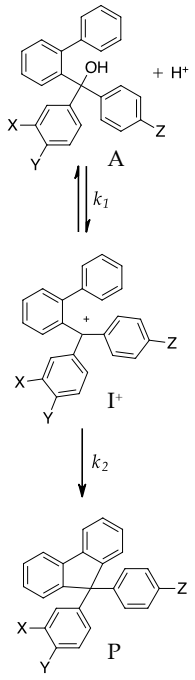


Table 1: Selected substituent constants. ⟵

A typographical error is present in the data table that was published. The relative value of k_{obs} for reactant **11** was reported to be 1.76 and the value that corresponds to the plotted data in Figure 1 of the paper is 0.176.[1] The correct value is used in this analysis.

Table 2: Selected substituent constants.↓

Substituent	σ	σ^+	σ^-
m-CF_3	0.43		
p-CF_3	0.54	0.61	0.65
m-OCH_3	0.12		
p-OCH_3	−0.27	−0.78	−0.26
m-CH_3	−0.07		
p-CH_3	−0.17	−0.31	
m-Cl	0.37		
p-Cl	0.23	0.11	0.19

The Hammett plot is presented in Figure 2. We see a classic example of a change in the rate-determining step. The authors plotted the four points describing a positive slope using σ^+ values and the eight points with a negative slope using σ values. As described above, I chose to use a consistent set of values.[2,3,4]

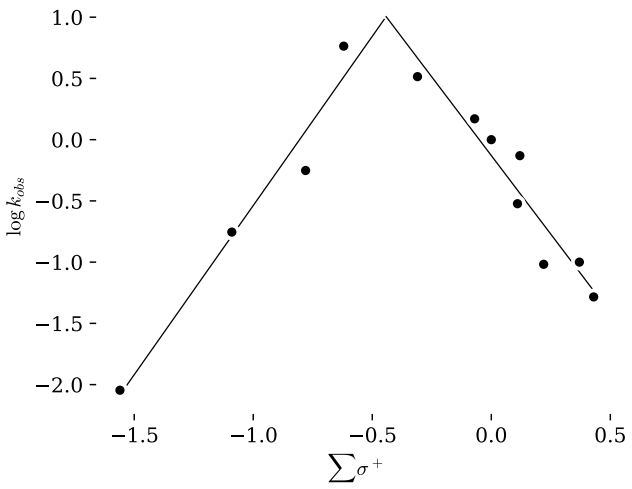


Figure 2: Hammett plot for cyclodehydration of substituted 2-phenyltriarylcarbinols.
←
The author's relative rate data was used with a correction as noted in the text. The substituent constants were recalculated using the more modern data set of Hansch, Leo & Taft. Only σ^+ values were used. Compare this plot to Figure 1 in the paper.[1]

[2] Should I have used σ rather than σ^+? Perhaps I should use a series of values where $\sigma' = \sigma + 0.70\,(\sigma^+ - \sigma)$. Recall that we have presented data for the hydrolysis of triarylcarbinol actates using all three of these sets of σ values in the previous two class meetings.

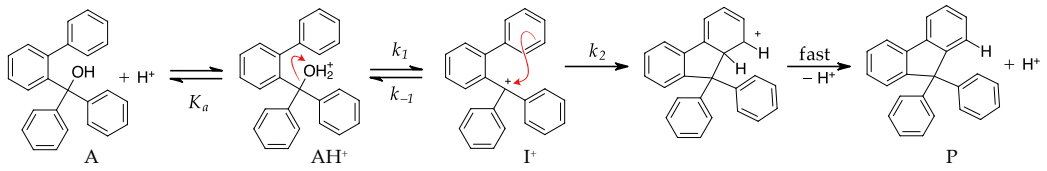
[3] "A survey of Hammett substituent constants and resonance and field parameters", C. Hansch, A. Leo, R.W. Taft, *Chemical Reviews*, **1991**, *91*, 165-195. https://doi.org/10.1021/cr00002a004. The authors used values from earlier work by Brown & Okamoto.

[4] The Hammett plot in this exploration was generated using the interactive *Python* notebook available via Google Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_16/CyclodehydrationNotebooks/HammetPlot.ipynb.

The Mechanism

The mechanism has two steps. First, a tertiary alcohol dissociates in acid to give the carbocation. Second, the cation forms a bond to an aromatic ring via electrophilic substitution. The curved Hammett plot is dominated by the slower step in each region. The step-by-step mechanism is outlined in Figure 3

Figure 3: The full proposed mechanism of the reaction.↓



The first step is an example of specific acid catalysis. Sulphuric acid is a very strong acid, and a significant amount of the protonated alcohol will exist at equilibrium. Water is a poor nucleophile, but it is a good leaving group. The triaryl carbocation intermediate is one of the most stable carbocations.

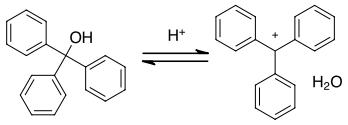
A carbocation is a strong electrophile. In this case, the strong electrophile is held in close proximity to a weak nucleophile (the electron-rich aromatic ring of the biphenyl group.) The intramolecular electrophilic addition to the aromatic system is rapid. The subsequent deprotonation to recover the

aromatic system is rapid. A protonated aromatic ring is one of the strongest acids in organic chemistry.

So we have a carbocation intermediate and a subsequent intramolecular Friedel-Crafts alkylation. In your organic chemistry experience, you will have been taught that formation of a carbocation is a rate-determining step, and you will also have been taught that electrophilic addition to a benzene ring is a rate-determining step. So, which is the r.d.s. here? It appears to change depending on the substituents.

The formation of a cation from a neutral starting material will have a negative slope in a Hammett plot. We see that, when the substituents are electron-withdrawing, the formation of the intermediate is the slow step. The slope of the Hammett plot is $\rho = -2.56$. The authors of the paper note that previous work has established that the reaction constant for the equilibrium dissociation of triaryalcohols is $\rho = -3.6$. This is in the same ballpark as the experimental results here. In the transition state, the bond is not fully broken and the ρ value for the rate should be less than in the fully dissociated case of the equilibrium.

Figure 4: Equilibrium for the dissociation of triarylcarbinol in strong acid. The reaction constant has been reported to be $\rho = -3.6$ (see reference 10 in the paper.[1])↓



The electrophilic addition to the biphenyl ring will result in the carbon connected to the substituted rings becoming neutral. The high-energy intermediate is also a carbocation, but the charge is now two atoms further away from the substituted rings. The charge on the central carbinol carbon atom has changed whom whole cationic to adjacent to a cationic atom. This is a negative change ($+1$ to 0 is a cahnge of -1). This step has a ρ value of 2.76. This unique situation has resulted in a system where we can change the r.d.s. depending on the substituents in the reactant.

Reaction Kinetics

Reaction kinetics can provide some information to help us interpret these results. Let us take the reaction as the following scheme.

$$\mathrm{A} + \mathrm{H}^+ \underset{\mathrm{K}_a}{\rightleftharpoons} \mathrm{AH}^+ \underset{\mathrm{k}_{-1}}{\overset{\mathrm{k}_1}{\rightleftharpoons}} \mathrm{I}^+ \xrightarrow{\mathrm{k}_2} \mathrm{P}$$

From this scheme, we can derive the following rate law.

$$\nu = \frac{[H^+]}{K_a + [H^+]} \frac{k_1}{k_{-1} + k_2} k_2 [A]_t$$

In the case where k_1 is the r.d.s., we say that $k_2 > k_{-1}$. If that is true, then the rate law becomes…

$$\nu = \frac{[H^+]}{K_a + [H^+]} k_1 [A]_t$$

The activity of H^+ has been set in each experiment (4% H_2SO_4), and the only value that will change is the value of k_1 for each substituted reactant. The reaction produces a cationic intermediate, and we expect a large negative

value of the Hammett reaction constant, ρ. So the negative slope region in the curved plot describes the change in the value of k_1 as substituents are changed.

The value of k_1 is increasing as substituents become electron-donating (a more stable cation is easier to make). We also expect k_2 to be diminishing with the same changes (a more stable cation is a weaker electrophile). Eventually k_1 becomes so fast that it is no longer the r.d.s. If we were to state that k_2 is the r.d.s., then we say that $k_2 < k_{-1}$, and the rate law becomes…

$$\nu = \frac{[H^+]}{K_a + [H^+]} \frac{k_1}{k_{-1}} k_2 [A]_t$$

The k_1/k_{-1} term represents the K_{eq} for formation of the intermediate and the rate of reaction now involves k_2. As the value of k_2 is reduced by more electron-donating substituents, the observed rate will also slow. The positive slope in the curved Hammett plot represents the effect of the change in the value of k_2 as substituents are changed.

Acid Catalysis

Note that the rate law includes a term for the amount of protonated carbinol. The reaction scheme proposes that this is an equilibrium that precedes the r.d.s. That would be specific acid catalysis. Can we confirm this? We generally do not use buffers in strongly acidic systems, so we cannot change the concentration of a buffer to see if the rate increases.

To determine which region of the Hammett plot is dependant on acidity, we will plot the rate vs acidity for each of the reactants to determine the reaction order in $[H^+]$. When protonation is part of the r.d.s. (either specific or general), we should see a first-order dependence in acid activity. In the reactants that have the aromatic substitution as the r.d.s., we should see no dependence of acid activity.[5]

[5] The acid activity plots in this exploration were generated using the interactive *Python* notebook available via Google colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_16/CyclodehydrationNotebooks/AcidRateOrder.ipynb.

Figure 5: Plot of k_obs vs H_0 for a series of reactants.
←

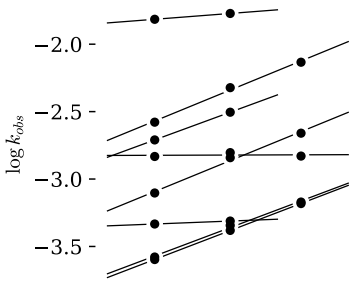


Table 3: Table of literature H_0 values and k_{obs} values for selected reactants.↓

% H_2SO_4	Literature H_0 values			k_{obs} for selected reactants $/10^{-4}s^{-1}$							
	HOAc[3]	Water[5]	70% DMSO[6]	**1**	**3**	**4**	**5**	**6**	**9**	**10**	**11**
4%	-2.38	0.20	0.06	2.53	2.64	7.88	19.5	26.4	14.7	153	4.64
5%	-2.46	0.09	-0.10	4.16	4.53	14.4	31.3	47.6	15.7	169	4.88
6%	-2.54	-0.02	-0.25	6.58	6.75	21.88		73.5	14.8		

The authors report their observed rates at three concentrations of sulphuric acid, 4, 5 and 6% $^{w}/_{w}$ H_2SO_4 in 80% aqueous acetic acid. To plot rates vs acidity we need a scale that measures acidity, not grams. The authors should have used an indicator dye to determine the H_0 of each mixture. In the absence of that information, I searched the literature looking for reported H_0 values of mixtures of 80% acetic acid and H_2SO_4. I found none (but I'm not done looking.) I did find a study of H_0 for mixtures of H_2SO_4 and glacial acetic acid.[6] I used the data and interpolated H_0 values for each of the three acid conditions. I then used the data reported in table IV of the paper.[1] The data and H_0 values from literature sources are collected in table 3.

The rates were plotted against H_0 values for each reactant. We see in Figure 5 that the reactants group into two classes. Those that should have a dependance on acidity and those that are independent of acidity. For the reactants in which k_{obs} is dependent on H_0, we see a slope in the log-log plot of approximately 2.5. This is greater than second-order. The H_0 values were for mixtures of pure acetic acid and H_2SO_4. But, the authors performed their experiments in 80% aqueous acetic acid. I could find no H_0 data for that situation.

Hammett has reported that, in aqueous acetic acid, the value of H_0 tracked more closely to the values with water alone.[7] So I will use H_0 values for H_2SO_4/H_2O mixtures. Data is scant for lower concentrations of H_2SO_4 but I found a data set[8] and interpolated H_0 values. I also found a data set for H_2SO_4 in 70% DMSO,[9] which seems similar to the water fraction in 80% acetic acid. The log-log plot slopes were 2.0 and 1.3 when these values were used. Without accurate H_0 data for the mixtures used in the paper, we cannot draw any conclusions about the order of the reaction in [H^+]. I feel that the 70% DMSO H_0 values are closest to the answer. This mixture has a similar water fraction in a polar solvent to the case of 80% acetic acid. The absolute acidity will be very different, but the relative acidity across the three concentrations of H_2SO_4 should be similar. This set of H_0 values produced a slope that approached the expected value of 1.0. However, we must be cognizant of confirmation bias and accept that we are doing a bit of "hand waving" here. I expect that if we had directly measured the H_0 of the three concentrations of H_2SO_4 using indicator dyes, we would see an order much closer to 1.

The order of reaction in [H^+] was calculated from the slopes in Figure 5. The slopes were determined against three sets of H_0 values as described above. The results are collected in table 4 and the plots of reaction order (n) vs $\sum \sigma^+$ are presented in Figure 6. We see that the order of reaction has a high magnitude when electron-withdrawing substituents are present and drops to near zero as the system becomes more electron-rich. This tracks with the change in r.d.s. The acid-catalyzed formation of the intermediate (k_1) should have first-order dependance ($n = 1$) in [H^+],[10] although the observed values are greater than this. What is clear is that the dependence of rate on [H^+] drops to zero when the electrophilic aromatic substitution step (k_2) becomes the r.d.s.

[6] "The Acidity Scale in Glacial Acetic Acid. I. Sulfuric Acid Solutions. $-6 < H_0 < 0$", N.F. Hall, W.F. Spengeman, *J. Am.Chem. Soc.*, **1940**, *62*, 2487-2492. https://doi.org/10.1021/ja01866a062

[7] "Acid Catalysis in Sulfuric Acid-Acetic Acid Solutions. The Rate of Bromination of m-Nitroacetophenone", M.A. Paul, L.P. Hammett, *J. Am. Chem. Soc.*, **1936**, *58*, 2182-2187. https://doi.org/10.1021/ja01302a027

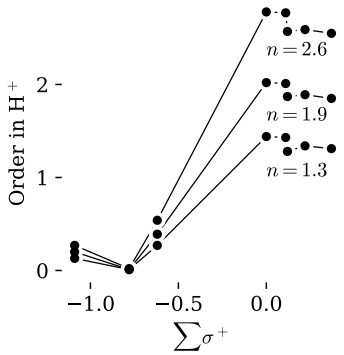
[8] "Temperature variation of the H_0 acidity function in aqueous sulfuric acid solution", C.D. Johnson, A.R. Katritzky, S.A. Shapiro, *J. Am. Chem. Soc.*, **1969**, *91*, 6654-6662. https://doi.org/10.1021/ja01052a021

[9] "Acidity Function Behavior in Aqueous-organic Solvents. I. The H_0, and H_A Functions in 70% DMSO/H_2O", K. Yates, G. Welch, *Can. J. Chem.*, **1972**, *50*, 1513 - 1522. https://doi.org/10.1139/v72-239

Table 4: Order of reaction in [H^+] measured by a plot of $\log k_{obs}$ vs H_0. Three solvent systems[3,5,6] that were deemed similar to the experiment are used for the H_0 values.↓

		Calculated Order of Reaction		
R	$\sum \sigma^+$	HOAc	Water	70% DMSO
1	0.22	2.59	1.89	1.34
3	0.37	2.55	1.85	1.31
4	0.11	2.77	2.01	1.43
5	0.12	2.57	1.87	1.28
6	0	2.78	2.02	1.44
9	-0.78	0.02	0.01	0.01
10	-0.62	0.54	0.39	0.27
11	-1.09	0.27	0.20	0.13

Figure 6: Plot of order of reaction in [H^+] (n) vs $\sum \sigma^+$ using three different sets of H_0 values. ↓



[10] Assuming that $H_0 < K_a$, which is likely true in the aqueous solvent.

Isotope Effects

Solvent isotope effects may provide insight. In general acid-catalyzed reactions, the bond to the proton is breaking or forming (depending on your point of view.) This changing bond in the transition-state would have a lower force constant than the covalent bond to the proton on either the acid or the intermediate. We should observe a significant primary kinetic isotope effect of $^{k_H}/_{k_D} > 1.5$.

We will calculate the solvent isotope effect on the observed rate. If the acid catalysis is general, we should see a significant effect. If the catalysis is specific, we will see an inverse solvent kinetic isotope effect (as will be explained below). We will begin with step one, the acidity rate profile.

If the case is specific acid catalysis, we have an equilibrium. The bonds to the acid proton are fully formed in both the starting material and the product. Is there a difference in force constant? Probably, but not enough to matter. However, in water, there is another important factor – the acid is hydronium ion.

The vibrational energy levels for bond-strecthing in D_2O are lower compared to H_2O. In D_3O^+, the force constant for the stretch is smaller. That means the difference in energy levels for that stretch is also smaller between D_3O^+ and H_3O^+. If you make a diagram, you will see that the energy difference for protonation is greater in the case of D_3O^+, due to the difference in vibrational energy levels. That means that D_2O is a weaker base than H_2O and, conversely, D_3O^+ is a stronger acid.

The autodissociation constant for D_2O is $pK_{W_{D_2O}} = 14.957$.[11] That means that pure D_2O is neutral when $pD = 7.48$. We can expect to see about $10^{0.48}$-fold greater protonating power for pD value compared to the same pH value in water.[12,13] The solvent isotope effect results reported by the authors in Table V of the paper (reproduced in Table 5 here) show that the rate in D_2O is indeed increased by that amount for reactants with substituents that result in the formation of the intermediate being the rate-determining step.

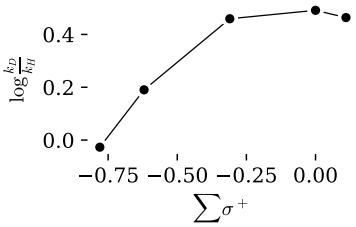
We observe that the values reported for $^{k_H}/_{k_D}$ are near $^1/_3$ for cases where the Hammett plot slope is negative (and where k_1 is the r.d.s.) This is exactly what we would expect ($10^{0.48} \simeq 3$). So we clearly see an inverse solvent kinetic isotope effect when the r.d.s. is the acid-catalyzed formation of the intermediate. For reactants that have a positive slope (and where k_2 is the r.d.s.) we see that the isotope effect approaches unity.[14]

From the pH-rate profile and the solvent kinetic isotope effect, we conclude that the formation of the intermediate is rate-determining when the substituents are electron-withdrawing and that this step is specific acid catalyzed.

Table 5: Solvent Isotope Experiment.↓

Reactant	$^{k_D}/_{k_H}$	$\sum \sigma^+$
4	2.91	0.11
6	3.10	0
8	2.88	-0.31
9	0.94	-0.78
10	1.55	-0.62

Figure 7: Plot of solvent isotope effect vs $\sum \sigma^+$. A significant isotope effect exists when substituents are electron-withdrawing and vanishes as the reactant becomes electron-rich.↓



[11] Shoesmith, D.W.; Lee, W., "The ionization constant of heavy water (D_2O) in the temperature range 298 to 523 K." *Can. J. Chem.*, **1976**, *54*, 3553-3558. https://doi.org/10.1139/v76-511.

[12] "The Deuterium Isotope Effect", K.B. Wiberg, *Chem. Rev.*, **1955**, *55*, 713-743. https://doi.org/10.1021/cr50004a004

[13] "Application Of The H_0 Acidity Function To Kinetics And Mechanisms Of Acid Catalysis", F.A. Long, M.A. Paul, *Chem. Rev.*, **1957**, *57*, 935-1010. https://doi.org/10.1021/cr50017a003

[14] The isotope effect plot in Figure 7 was generated using the interactive *Python* notebook available via Google colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_16/CyclodehydrationNotebooks/IsotopeEffect.ipynb.

Thermodynamics

Erying plots of rate vs temperature experiments will reveal the changes in entropy and enthalpy of the transition state for the r.d.s. The authors report such data in table VII in the paper and the results of the Erying plots in table I of the paper[1]. There is a large change in $\Delta S^{\ddagger}$ as the r.d.s. changes. However, I noted a discrepancy for the most electron-rich reactant (**12**) in the calculated results for $\Delta S^{\ddagger}$ presented in table I. I will use this data to calculate values for $\Delta H^{\ddagger}$ and $\Delta S^{\ddagger}$ and check to see if there is a typographical error or not. The data is presented in table 6. The Erying plots are presented in Figure 8.

You can see that there are some data series with three or two points. In the paper, the authors added back in data points by "extrapolating from Arrhenius plots." I presume that they plotted $\ln k_{obs}$ vs $1/T$ and then used the pre-exponential factor and energy of activation to calculate rates at the temperatures missing data. They then plugged these values into the Erying plot. Making up data points by extrapolating a two-point plot does not actually get you more data points. You still have two actual points. Obviously, I disagree with this approach and will not be using it. The data in table 6 in this analysis presents only the actual data.

Table 6: Data from table VII of the paper.[1]↓

Temp	k_{obs} for Reactants/$10^{-4}s^{-1}$											
/degreeCelsius	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
15				1.87	6.39	9.66	12.3	23.3	5.6	53.4		
20	1.26		1.54	3.90	9.88	15.5	21.8	44.5	8.82	99.2	2.79	
25	2.53	1.36	2.64	7.88	19.5	26.4	39.2	86.3	14.7	153	4.64	0.238
30	4.93	2.25	5.06	12.2	33.0	50.4	66.2	137	24.4		8.00	
35	8.89	13.5	9.94	23.8		76.2			36.0			0.701

The Erying equation relates rate to temperature using transition state theory.

$$k_{obs} = \frac{\kappa k_B}{h} T \cdot e^{\Delta S^{\ddagger}/R} \cdot e^{-\Delta H^{\ddagger}/RT}$$

This can be plotted in the linear form as…

$$\ln \frac{k_{obs}}{T} = \frac{-\Delta H^{\ddagger}}{R} \frac{1}{T} + \ln \frac{\kappa k_B}{h} + \frac{\Delta S^{\ddagger}}{R}$$

$\kappa = 1$
$k_B = 1.380649 \times 10^{-23}\ J \cdot K^{-1}$
$h = 1.054572 \times 10^{-34}\ J \cdot s$
$R = 8.314463\ J \cdot mole^{-1} K^{-1}$

Plotting $\ln \frac{k_{obs}}{T}$ vs $\frac{1}{T}$ should result in a straight line with a slope equal to $\frac{-\Delta H^{\ddagger}}{R}$ and a y-intercept of $\frac{\Delta S^{\ddagger}}{R} + \ln \frac{\kappa k_B}{h}$.

From the Erying plot in Figure 8, we can see that one data set is clearly out of step with the rest. The data for reactant **2** has an unexpectedly high value for k_{obs} at the highest temperature. Is there a typographical error here? In general, my values for $\Delta H^{\ddagger}$ and $\Delta S^{\ddagger}$ were consistent with the results reported

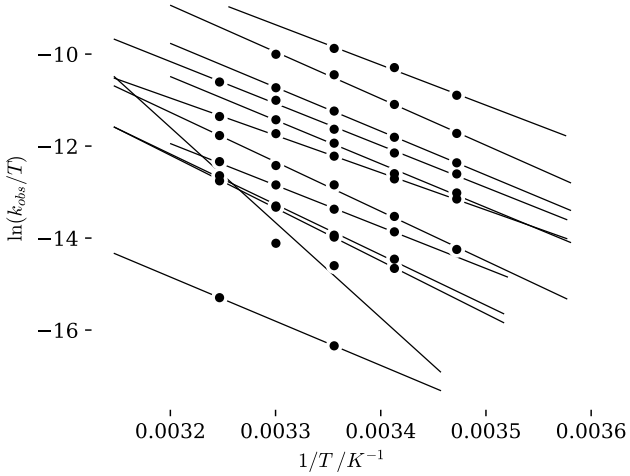


Figure 8: Erying plot of experimental data from table VII of the paper.[1]
⟵

in the paper, but there were significant differences. My own interpretation of that data does not show a sudden change in the value for $\Delta H^{\ddagger}$ when the change in r.d.s. is believed to occur, but there is a trend in that direction. The values reported by the authors showed a very clear change. There is no methodology discussed for how curve fits were performed and no documentation of calculations[15] provided by the authors.

I also performed the Erying plot using the data set provided by the authors, which includes the "extrapolated" points, and I did not obtain their values (but was closer). The data for reactants **1** and **2** provided very different answers from those reported by the authors. I now suspect that there was a typographical error in the extrapolated data for reactant **1** as well as in the experimental data for reactant **2**.

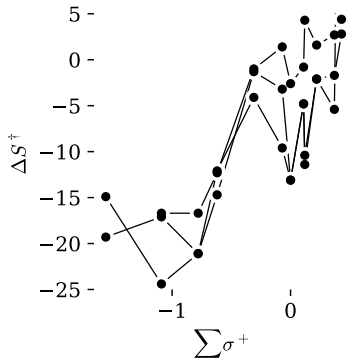
[15] See the accompanying *Jupyter* notebook for documentation on my own calculations

The value for $\Delta S^{\ddagger}$ changes according to the r.d.s. The dissociation step involves separating a group from the molecule. Generally, splitting a molecule has a positive value for $\Delta S^{\ddagger}$, but in aqueous systems, there can be other effects. In the rate-determining step, the departing water is becoming solvated and that creates more order in the solvent as hydrogen bonds align. In the case of the values reported by the authors, the $\Delta S^{\ddagger}$ for the first step, k_1, is near zero. In the reactants where the aromatic addition is the r.d.s., we see that $\Delta S^{\ddagger}$ is a significantly negative value. This is expected in ring-closing as we are locking out bond rotations with the ring formation.

My own results using the experimental data only were not as clear cut but still show large negative values where k_2 is the r.d.s. These results and the results calculated from the authors' data with "extrapolated" points are presented in table 7, along with the reported results from table I in the paper.

I plotted these $\Delta S^{\ddagger}$ values against the $\sum \sigma^{+}$ value for each reactant as shown in Figure 9. You can see the trend in $\Delta S^{\ddagger}$ with larger negative values

Figure 9: Plot of $\Delta S^{\ddagger}$ values against the $\sum \sigma^{+}$ for the results reported in the paper, results calculated from the edited "extrapolated" data and results calculated from experimental data only.↓



	$\Delta S^{\ddagger}$ values			
Reactant	$\sum \sigma^{+}$	Extrapolated	From Exp.	Table I
1	0.22	-2.1	-2.1	1.6
2	0.43	2.8	**57.8**	4.4
3	0.37	-1.7	-5.4	2.7
4	0.11	-4.8	-4.8	-0.8
5	0.12	-10.4	-11.4	4.3
6	0	-13.1	-13.1	-2.6
7	-0.07	-3.2	-9.6	1.4
8	-0.31	-1.3	-4.1	-1.0
9	-0.78	-21.1	-21.1	-16.7
10	-0.62	-14.7	-12.1	-12.3
11	-1.09	-24.4	-17.1	-16.7
12	-1.56	-14.9	-19.3	**-4.9**

Table 7: Data from Erying plots and Table I of the paper.[1]
⟵
"Extrapolated" is $\Delta S^{\ddagger}$ values calculated using Erying plots of the values reported in table VII of the paper. "From Exp" used only the experimental data from table VII. "Table I" is the $\Delta S^{\ddagger}$ values calculated by the authors and reported in table I of the paper. The values in bold were deemed to be likely erroneous and were not included in the plots presented in Figure 9.

when the r.d.s. is k_2.

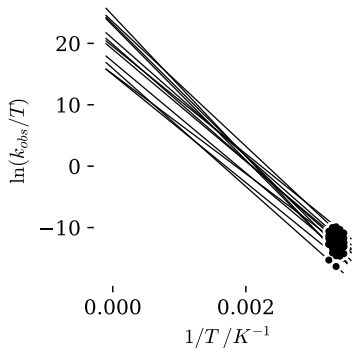
The authors experiment was set over a narrow range of five temperatures. Some reactants only had two data points to make the line. I would recommend at least seven data points for Eyring plots – and make every effort to maximize the accuracy of your observations. Precise temperature control is required along with careful measurements.[16]

Consider if we view the Erying plots in Figure 8 at a scale that includes the y-intercept as shown in Figure 10. You can see how far the "dart is thrown" for each curve fit. Small imperfections in the data can change the intercept value significantly. Many researchers consider the Erying plot unreliable for determining $\Delta S^{\ddagger}$ because of the long-range extrapolation (we are actually extrapolating to infinite temperature and that is a pretty big jump)[17]

There is another linearization of the Erying equation that reveals $\Delta S^{\ddagger}$ as the slope, not the intercept. One's instincts say that the slope of five points (a kind of average, really) will be more precise than an extrapolation to the intercept. However, it makes little difference. A mathematical proof of this statistical issue has been published.[18] In the end, the quality of the data is the most important factor.

[16] The Eyring effect plot in this exploration was generated using the interactive *Python* notebook available via Google colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_16/CyclodehydrationNotebooks/EryingPlots.ipynb.

Figure 10: The Eyring plot from Figure 8 scaled to include the y-intercept. ↓



[17] How far away do you stand from your dartboard and how reliable is your aim?

[18] "A common misconception about the Eyring equation", G. Lente, I. Fábián, A.J. Poë, *New J. Chem.*, **2005**, *29*, 759-760. https://doi.org/10.1039/B501687H

Conclusion

The authors have provided evidence that supports their hypothesis that the cyclodehydration of 2-phenyltriarylcarbinol is a reaction that is specific acid catalyzed and that the rate-determining step can change with changing electron density in the aryl rings. The Hammett plot, H^{+} ion molecularity, and solvent kinetic isotope effect all reflect this. Thermodynamic analysis of the kinetics provides the same conclusion, although with less certainty.

Several typographical errors were found or are suspected. The kinetic data

does not give the same results as those calculated by the authors (although they are similar) and no documentation as to the computational methods was available to evaluate the reason for this.[19]

[19] See the accompanying *Jupyter* notebook for documentation on the data analysis presented in this report. You can spare your readers lots of boring-yet-essential details and provide them with the exact computational methods by using *Jupyter* notebook or similar tools to document your math and methods.

Exploration Tips

This is an example of the scope of work expected in a student exploration. Observe that I replotted the graphs and found typographical errors in the data tables. I also used the data to perform my own analysis of the order of reaction vs. acid activity and had to dig up some values for H_0 for sulfuric acid/acetic acid mixtures. I investigated the relationship between substituent and $\Delta S^{\ddagger}$ and confirmed the proposed change in mechanism. There was also an opportunity to show how the isotope effect shows the same information.

Look to how I went beyond the single publication for more information. Often, the authors will reference all the extra papers that you need, but sometimes you will need to search on your own. I would like a student exploration to include some evidence of digging into the data.